

Supplementary information for: Destruction of long-range order in non-collinear two-dimensional antiferromagnets by random-bond disorder

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I. SINGLE BOND DEFECT

Here we provide a few details concerning the physics of a single defect bond.

A. Spin texture via linear response

As explained in the main text, the spin texture induced by a bond defect can be understood as the response to a local dipolar transverse field which acts to rotate the two spins on the defect bond away from their clean-limit configuration. This is most efficiently calculated in a local frame where the clean-limit state is uniform.

To this end we start from a coplanar state in the $x-y$ plane and rotate the spins such that all point along the $\hat{x}$ axis. The rotation involves the y components of the spins, i.e., the susceptibility $\chi^\perp(\vec{q}) = \langle\langle S_{\vec{q}}^y; S_{-\vec{q}}^y \rangle\rangle$. For a system with underlying SU(2) spin symmetry, this susceptibility is governed by the in-plane Goldstone modes and generically behaves as $\chi^\perp(\vec{q}) \propto 1/q^2$ for small momenta. For the triangular-lattice Heisenberg model in the classical limit the full expression for the transverse susceptibility is given by [1]

$$\chi^\perp(\vec{q}) = S \frac{1}{\gamma(\vec{q}) - 1} \quad (\text{S1})$$

where the triangular lattice structure factor

$$\begin{aligned} \gamma(\vec{q}) &= \frac{1}{6} \sum_{\vec{\delta} \in \text{NN}} e^{i\vec{q} \cdot \vec{\delta}} \\ &= \frac{1}{3} \left(\cos q_x + 2 \cos \frac{q_x}{2} \cos \frac{\sqrt{3}}{2} q_y \right) \end{aligned} \quad (\text{S2})$$

encodes the geometry of the lattice. For a bond defect at a position $\vec{r}'$ on a coupling along the direction $\vec{\delta}(\vec{r}')$, the transverse response at a separate site $\vec{r}$ is given by

$$\langle S_{\vec{r}}^\perp \rangle = \sum_{j=0}^1 h_{\vec{r}}^\perp \beta_j \langle\langle S_{\vec{r}}^y; S_{\vec{r}'+\beta_j \vec{\delta}(\vec{r}')/2}^y \rangle\rangle \quad (\text{S3})$$

where $\beta_j = (-1)^j$ is the dipolar form factor of the perturbation, and $h^\perp \propto \delta J/J$ measures the strength of the perturbation. The proportionality factor, which we call

$\tilde{\alpha}$, depends on the microscopic details of the lattice and in itself is proportional to the stiffness of the ordered phase. Later, we extract the values of this non-universal prefactor for two different non-collinear antiferromagnets, the triangular-lattice Heisenberg model and the J_1-J_4 cubic-lattice Heisenberg model.

Once the transverse response is expressed in terms of the susceptibility $\chi^\perp(q)$,

$$\langle S_{\vec{r}}^\perp \rangle = \frac{\tilde{\alpha}}{J} \sum_{\vec{q}} \delta J(\vec{r}') \tilde{\beta}_{\vec{\delta}(\vec{r}')}(\vec{q}) \chi^\perp(\vec{q}) e^{i\vec{q} \cdot (\vec{r}-\vec{r}')} \quad (\text{S4})$$

with the momentum dependence of the form factor given by $\tilde{\beta}_{\vec{\delta}(\vec{r}')}(\vec{q}) = e^{-i\vec{q} \cdot \vec{\delta}(\vec{r}')/2} - e^{i\vec{q} \cdot \vec{\delta}(\vec{r}')/2} = -2i \sin(\vec{q} \cdot \vec{\delta}(\vec{r}')/2)$, its dipolar nature becomes apparent. The long wave length response,

$$\langle S_{\vec{r}}^\perp \rangle = \frac{\alpha S}{J} \sum_{\vec{q}} \delta J(\vec{r}') \frac{\vec{q} \cdot \vec{\delta}(\vec{r}')}{q^2} e^{i\vec{q} \cdot (\vec{r}-\vec{r}')} \quad (\text{S5})$$

and thereby the spin texture of the resulting state is determined by the defect induced dipole moment $\vec{d}(\vec{r}') = (\delta J(\vec{r}')/J) \vec{\delta}(\vec{r}')$ and the quadratic dispersion of the Goldstone modes. Once again the prefactor α (see also in main text) depends on the underlying lattice and following (S2), for triangular lattice, we get $\alpha = 4i\tilde{\alpha}$.

B. Uniform moment

In the presence of a single defect bond, the bulk spin moments do not exactly cancel each other, such that the resulting state has a non-vanishing uniform magnetization m_{imp} . For small δJ this is proportional to δJ ,

$$m_{\text{imp}} = \gamma S \delta J + \mathcal{O}(S^0). \quad (\text{S6})$$

For the triangular-lattice Heisenberg model we numerically find $\gamma = [\text{ToDo: add number}]$, see Fig. [\[ToDo: add fig\]](#). The proportionality can be expected to break down for larger δJ , but still holds to very good accuracy for $\delta J = -J$ where we find $m_{\text{imp}} = 0.41S + \mathcal{O}(S^0)$. [\[ToDo: can you get the 1/S correction?\]](#)

A finite-size scaling analysis of m_{imp} is shown in Fig. [\[ToDo: add fig\]](#).

II. DESTRUCTION OF LONG-RANGE ORDER

As explained in the main text, an estimate of the transverse spin fluctuations induced by a collection of bond defects can be used to assess the destruction of long-range order.

In the linear response regime, we can consider the superposition of individual dipolar responses (S5) from a spatially random distribution of bond defects. The average transverse response is simply zero for a defect distribution with zero mean and therefore their collective effect is manifested through the squared transverse fluctuation. Although the angular profile of the response due to each bond defect is dependent on the orientations of the respective bonds, for the case of a perfectly random distribution of such defects, $\delta J(\vec{r})\delta J(\vec{r}') = (\delta J)^2 \delta_{\vec{r}, \vec{r}'}$, the contribution coming from the separate bond or dipolar orientations decouple. From (S5) it is also easy to see that these contributions differ by phase factors of constant angular rotations. Therefore by choosing appropriately rotated frames of axes for each, we can cast them in an identical form and sum up to find the expression for the total transverse spin fluctuation,

$$\begin{aligned}
\overline{\langle S_0^{\perp 2} \rangle} &= \left(\frac{\alpha}{J} \right)^2 \sum_{\vec{r}, \vec{r}'} \sum_{\vec{q}, \vec{q}'} \overline{\delta J(\vec{r})\delta J(\vec{r}')} \\
&\quad (\vec{q} \cdot \vec{\delta}(\vec{r}))(\vec{q}' \cdot \vec{\delta}(\vec{r}')) \chi^\perp(\vec{q}) \chi^\perp(\vec{q}') e^{i(\vec{q} \cdot \vec{r} + \vec{q}' \cdot \vec{r}')} \\
&= \left(\frac{\alpha \delta J}{J} \right)^2 \sum_{\vec{r}} \sum_{\vec{q}, \vec{q}'} (\vec{q} \cdot \vec{\delta}(\vec{r}))(\vec{q}' \cdot \vec{\delta}(\vec{r})) \\
&\quad \chi^\perp(\vec{q}) \chi^\perp(\vec{q}') e^{i(\vec{q} + \vec{q}') \cdot \vec{r}} \\
&= \left(\frac{\alpha \delta J}{J} \right)^2 \frac{1}{N_B} \sum_i \sum_{\vec{q}} (\vec{q} \cdot \vec{\delta}_i)(\vec{q} \cdot \vec{\delta}_i) \chi^\perp(\vec{q}) \chi^\perp(\vec{q}') \\
&= \left(\frac{\alpha \delta J}{J} \right)^2 \int \frac{d^d q}{(2\pi)^d} (q_x \chi^\perp(\vec{q}))^2
\end{aligned} \tag{S7}$$

where $\chi^\perp(q) \propto S/q^2$ is the transverse susceptibility from the previous section and N_B is the number of distinct bond orientation for a given lattice geometry. It is noteworthy that the additional factor from the summation over the bond orientation is cancelled by the overall volume fraction for each such contribution. As discussed in the foremost part of this letter, if the susceptibility $\chi^\perp(\vec{q})$ is dominated by massless Goldstone modes, the integral (S7) diverges below a lower critical dimension. This signals the destruction of the long-range order and precisely at the critical dimension emergence of exponentially growing domains. In the following we consider the criterion $\overline{\langle S^{\perp 2} \rangle} \simeq S^2$ and provide a detailed supplement to our original discussion on various aspects of the destruction of long-range order in bond-disordered non-collinear magnets.

A. SU(2) symmetry and $T = 0$

For spontaneously broken SU(2) symmetry in a magnetic system the fundamental massless excitations have quadratic dispersions and therefore the transverse susceptibility $\chi^\perp(\vec{q}) \propto 1/q^2$. The convergence criterion of the integral (S7) makes the correlation length of the order parameter short-ranged,

$$\left(\frac{\alpha S \delta J}{J} \right)^2 \int_{1/\xi}^1 q^{d-1} dq \left(\frac{1}{q} \right)^2 \simeq S^2 \tag{S8}$$

This implies a correlation length $\xi^{2-d} \propto (J/\delta J)^2$ for $d < 2$, while for $d = 2$, barring the non-universal stiffness dependent factor α

$$\xi \propto e^{(J/\delta J)^2}. \tag{S9}$$

B. Finite T

At finite T and at $d = 2$ the magnet would be thermally disordered, but similar to the effect of bond disorder from the previous segment is exhibits exponential correlation growth, $\xi \propto e^{1/T}$ [2]. As temperature decreases, the thermal and quenched bond disorder competes for precedence and for a particular value of disorder strength a crossover temperature emerges, $T^* = \delta J/J$. If the temperature decreases any further the disordered state becomes magnetic. **[ToDo: This is apparently contentious now, so we should change. In Chakravorty Halperin I see a factor of spin-stiffness in the exponent, should we try to be precise and try get an appropriate cancellation of this with our prefactor α ?**

C. Layered systems

A realistic layered system displays a small inter-layer coupling which tends to stabilize magnetic order at finite temperatures in the presence of SU(2) symmetry. We assume an (unfrustrated) inter-layer coupling of magnitude $J_\perp$. Using the ratio between the interplanar and intraplanar couplings, $\kappa = J_\perp/J$ we can express the long-wavelength transverse susceptibility of the coupled layered system as

$$\chi^\perp(\vec{q}, q_\perp) = \frac{\lambda S}{q^2 + \kappa^2 q_\perp^2} \tag{S10}$$

where as before we have a dimensionless lattice dependent factor λ . Substituting this expression in the integral relation (S7) it becomes apparent that only the impurities among the in-plane bonds affect the long-range order. The stability criterion is determined by the coupling

ratio κ alone,

$$\left(\frac{\lambda\alpha S\delta J}{J}\right)^2 \int \frac{d^2 q dq_\perp}{(2\pi)^3} \frac{q_x}{q^2 + \kappa^2 q_\perp^2} \leq S^2 \quad (\text{S11})$$

with long-range ordered phase stable if,

$$\frac{J_\perp}{J} = \kappa \geq \frac{1}{2} \left(\frac{\lambda\alpha\delta J}{J}\right)^2 \quad (\text{S12})$$

D. Broken SU(2)

For a system with broken SU(2) symmetry, some or all of the Goldstone modes acquire a gap. For simplicity we assume a particular form of the response function,

$$\chi^1(\vec{q}) = \frac{\lambda S}{q^2 + \Delta^2} \quad (\text{S13})$$

with the dimensionless phenomenological factor λ absorbing the particular details of the lattice.

III. INCOMMENSURABILITY IN ANISOTROPIC HEISENBERG MODEL

We consider a spatially anisotropic Heisenberg model on the triangular lattice, with couplings $J_{1,2,3}$ on the bonds 1,2,3, with mean J . In the classical limit, the

ground state is given by a single- Q coplanar spiral whose propagation wavevector $\vec{Q}$ can be found by minimizing the Fourier-transformed coupling constant $J(\vec{q})$. In the isotropic case, the system orders at $\vec{Q}_0 = \pm(4\pi/3, 0)$.

The degree of anisotropy can be parameterized by $\vec{d} = \sum_{i=1}^3 (J_i/J) \hat{e}_i$ where $\hat{e}_i$ are the directed unit vectors along the nearest-neighbor bonds of the lattice. As explained in the main text, $\vec{d}$ takes the role of a local dipole density in the bond-disordered case.

For small deviations from isotropy, $|\vec{d}| \ll 1$, the relation between $\vec{Q}$ and $\vec{d}$ can be determined analytically **[ToDo: work out, write down]**.

IV. NUMERICAL RESULTS

V. BOND DEFECT IN THREE DIMENSIONS

We have checked that our scenario for bond disorder applies to three space dimensions as well. To this end, we have considered a single bond defect in **[ToDo: describe model and results]**.

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- [1] A. Wollny, PhD Thesis. (2016).
 - [2] S. Chakravorty, B. I. Halperin, and D. R. Nelson Phys. Rev. Lett. **60**, 1057 (1988).
 - [3] A. Andreanov, J. T. Chalker, T. E. Saunders, and D. Sherrington, Phys. Rev. B **81**, 014406 (2010).